

even a 200 GB hard drive. Ditto for sound cards and speakers. And CD/DVD-Writers. How much RAM does everyone need? Do they all need to connect to the Internet? And let's not forget that apart from all this, you also have to throw in a file server!

So, then, if you have an "Accounts" group that consists of, say, 10 employees, you have 10 hard disks, 10 256 MB RAM sticks (for example), etc. Basically, everything is multiplied by 10—it's unnecessary, and a complete waste of money! Wouldn't it be a lot easier to maintain a server that has all your data and files that need to be accessed, with a huge amount of space and RAM, and give all 10 employees access to it with barebones PCs? Welcome to the world of thin clients!

Weight-Loss 101

The concept is simple: invest in a thin client setup from one of the many service providers, get a nice, robust server, and, in our hypothetical case, 10 thin client PCs that come with no hard drives, just enough RAM to give you fast access to anything (generally 256 MB DDR), and none of the unnecessary add-ons like sound cards or optical drives. These are, obviously, cheaper than full-fledged computers, and you will still need the basics—a monitor, keyboard and mouse.

The servers will set you back a pretty penny, of course, and will probably have dual-CPUs, at least 2 GB of RAM, and in some cases a terabyte of storage space. The difference is that this server will have all the necessary applications loaded, and will be the only thing your system's admin will need to worry about. None of the 10 accounts employees in our example will have any hard drives or software loaded on their computer. All they have is, say, 512 MB of Flash memory that holds the basic OS (Linux or Windows CE/XP embedded), and everything they need to get the PC up and running and connected to the server. If your company uses, say, Tally, all employees with access to the accounting software loaded on the server will be able to run it simultaneously—from the server—after logging in.

Now the same goes for, say, the sales team. They may need access to Microsoft Office and e-

Thin, Thick, Fat, Huh?

Thin clients come in various flavours. Some of them have embedded OSes, and have everything a normal desktop would have, except for a hard drive. These are Thick clients. In such cases, all the processing is done by the client, and the server holds just the data and the applications.

Other thin clients are really thin, and come only with a basic processor and RAM, and enough hardware to connect you to a network. All processing and data storage is done by the server.

Then there are solutions that will let you use your old hardware, which you would get rid of when you upgraded, to act as thin clients.

Then there are the peripherals which fit into fully functional desktops, allowing a few more monitors and input devices to be connected, so that you have one PC and between two and four employees using it simultaneously. These are not really thin clients, as you may have guessed, but do the job just as efficiently for small businesses with less than 15 computer users—just buy four or five powerful computers and connect everyone with additional keyboards and monitors!

mail—and nothing else. So you set the necessary permissions on the server, and all the employees who make up the "Sales" group have access to whatever software you allow for that group.

Of course, let's not forget that you could also use Linux in place of Windows, and save even more money.

Skinny Supermodels

There are quite a few companies that are providing thin client hardware, software and services. The most notable around the world are Wyse (www.wyse.com), Chip PC (www.chippc.com), HP (www.hp.com), and Sun Microsystems (www.sun.com).

All these manufacturers make very elegant-looking thin client workstations. Chip PC needs special mention here for making some really *small* thin clients. An example is their Jack PC, which can actually fit into a standard LAN jack wall mounted box—those little white boxes that we connect LAN jacks to in offices!

Lean, Mean, Working Machines?

It's very obvious from the above that thin clients will save a company a lot of money, but what about security and uptimes? These were the first two needs for any business that we listed. Well, consider the fact that in a thin client environment, there really is nothing to manage except the server(s). So all a system admin needs to do is make sure the network stays up, and that the server remains secure and runs. And it's certainly a lot easier to manage the security of data at one location than at several.

Depending on how thin your clients are (whether they run an OS themselves, or boot from the LAN), there's virtually no risk from viruses and worms, except to the server, of course, which had better be well protected.

All companies with over 50 computers in their office should consider thin computing as a viable alternative. All you need to do is contact one of the companies we mentioned, and they'll help you choose thin clients that are ideal for your requirements. ■

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Chip PC's Jack PC is only as big as a wall-mounted LAN jack board. Now that's thin!